

Measuring emergence through collapse of the effective joint possibility space

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Abstract

Emergence is invoked across physics, biology and machine learning, yet each field certifies it with a different instrument, so identical macroscopic behaviours can be labelled emergent or not depending on the measure chosen. Here we define emergence as a spontaneous, persistent, regime-level collapse of a system’s effective joint state–action–trajectory possibility space, decomposed by source into environment-mediated, individual, pairwise and higher-order channels. Using a preregistered protocol, we calibrate the instrument on analytic ground truth, validate it on off-design physics, and apply it to learned multi-agent systems. An onset-type breakpoint in the collapse appears where a system commits to a new joint regime—including in learned populations with no designed commitment gate—and not under ordinary smooth optimization; its timing and sharpness shift systematically with system size and distance from criticality, and remaining openness predicts whether an intervention can still change the outcome. This provides a calibrated, falsifiable operationalization of a contested word.

Introduction

“More is different” is a claim about organization: many interacting parts settle into a collective regime that none of the parts specified in advance^{1–4}. The word *emergence* names that phenomenon in statistical physics^{5,6}, in the self-organization of animal collectives^{7–10}, in the appearance of new capabilities in large neural networks^{11,12}, and in multi-agent reinforcement learning^{13–15}. Yet the fields do not share an instrument. Integrated-information and causal-emergence measures ask whether a coarse-grained description has more causal power than its microstate^{16–19}; information-theoretic autonomy measures test for macro-variable self-determination^{20,21}; partial-information decomposition asks how much of a target is carried synergistically^{22,23}; phase-transition theory tracks an order parameter²⁴; complexity science invokes computational and statistical irreducibility^{25,26}; and the machine-learning literature reports “emergent abilities” as sharp upturns of a benchmark against model scale^{11,27,28}, a debate entangled with scaling laws^{29,30} and with grokking, where generalization appears long after memorization^{31,32}.

These instruments disagree in a way that is not cosmetic. A recent and influential critique showed that some reported emergent abilities are artefacts of discontinuous metrics: change the score function and the sharp jump disappears³³. This is not an isolated embarrassment; it is the generic risk of defining emergence on the *observable of convenience*—a benchmark, an order parameter, a success rate—rather than on the object that actually reorganizes. Two systems can produce byte-identical macroscopic

outcomes—identical joint distributions, identical marginals, identical task success—through entirely different underlying couplings (Fig. 1a), and no statistic of that shared distribution can tell them apart. If “emergence” is read off the outcome, these systems are permanently indistinguishable. They become distinguishable only relative to a declared *observation contract*—which variables are measured, which are exogenous, and which interaction-severing counterfactuals are admissible—under which one can ask how the effective possibility space closes. That is what our instrument measures; it attributes structure under a stated contract rather than claiming to recover causal mechanism from outcome statistics alone.

We take the second view and make it operational. Consider the joint space of what every part *could* still do together—the effective joint state–action–trajectory space. Before organization this space is wide; a colony of ants approaching a gap can still form a bridge, cross elsewhere, or disperse³⁴. Emergence, on our account, is the moment this space *collapses* onto a narrow committed regime: the colony decides, together, to build. Crucially the collapse is not the completion. The bridge does not yet exist when the collective possibility space has already contracted onto building it. This is the intuition our instrument is built to measure, and to measure in a way whose verdicts are demonstrably stable within a measured admissible equivalence class of contracts—reference space, truncation, window, sampling grid and representation (Methods)—rather than inheriting the fragility of the observable of convenience.

Here we make three contributions. First, a definition with a two-level structure: a qualification (spontaneous, regime-level, persistent) and a quantitative *emergence intensity profile* (EIP) whose axes—amplitude, abruptness, and the source decomposition—are separately measured, so that abruptness is a phenotype rather than a smuggled existence criterion. Second, a calibrated measurement contract, matured through preregistered self-falsification, that recovers the source of a collapse on analytic ground truth (72/72 cells) and on physics the instrument was never designed around. Third, and most consequential for this journal’s readers, the finding that in *learned* systems the same collapse admits a clean dissociation of timescales, varies systematically and reproducibly with system size and coupling, predicts controllability, and—importantly—is **not** generic: the punctuated form arises spontaneously where learning must cross a joint exploration barrier (convention formation, role lock-in), while ordinary dense-reward multi-agent optimization collapses smoothly, and we report exactly where it does and does not appear.

Results

An operational definition on the right object

We measure openness as the normalized entropy of the effective joint possibility distribution, $O_t = H(P_t)/H(P_{\text{ref}})$, and collapse as $C_t = 1 - O_t$. Because literal support never shrinks for a stochastic policy, we work throughout with the *effective* possibility count $N_{\text{eff}} = 2^H$; this is the formal version of the $10^n \rightarrow 2^n$ intuition. A system *qualifies* as emergent when its collapse is regime-level (a structural change in the joint distribution, not a marginal one), endogenous (generated internally, declared through a provenance boundary rather than assumed), and persistent (the closed regime remains stable over the declared observation horizon; recovery from perturbation and causal irreversibility are reported separately as robustness properties, not folded into the definition). Given qualification, we report an intensity profile rather than a binary label. P_t itself is defined relative to a declared *observation contract*—which variables enter the joint space, over what horizon, at what discretization, against which reference,

with which provenance boundary—frozen before any adjudication; objectivity enters through freezing, measured invariance and refusal outside the contract’s domain, not through a claim of representation-free measurement (Methods).

The decisive move is the source decomposition. Total collapse splits along a nested maximum-entropy ladder,

$$C_{\text{total}} = C_{\text{env}} + C_{\text{individual}} + C_{\text{pair}} + C_{\text{high}}, \quad (1)$$

so that the interaction cut is a *decomposer*, not a gate, in the spirit of connected-correlation expansions³⁵. On analytic ground truth with four independent knobs, each knob moves only its own component, the ladder is monotone, and hiding an environmental common cause deterministically re-attributes collapse from C_{env} to the relational channels (all five preregistered checks pass; Fig. 1d). This contract-relativity is not a bug: it is the precise, measurable statement that “higher-order” is defined only relative to what an observer declares exogenous. On a full factorial of 72 analytic cells (4 sources $\times$ 3 temporal shapes $\times$ 2 stabilities $\times$ 3 magnitudes) the instrument recovers the source in 72/72 cells, and—the point that defends us against the metric-artefact critique³³—the amplitude M is invariant across temporal shape (relative range 0.000 across all groups) while the abruptness J strictly orders punctuated $>$ sigmoid $>$ gradual (Fig. 1c). Amplitude and abruptness are measured as different quantities; a revelation-only or discontinuous-metric control produces exactly zero collapse.

A breakpoint marks the punctuated phenotype, not emergence itself

Qualification distinguishes emergent from non-emergent organization; the *breakpoint* distinguishes the punctuated profile from gradual, non-monotone or unresolved collapse—absence of a breakpoint therefore never by itself rules emergence out. We fit the collapse curve with a two-regime model and compare it to a one-regime fit by ΔBIC , inside a saturation-truncated window and behind an effect-size gate—two amendments we were forced to adopt when a flat, subcritical control produced a spurious hinge and when an S-curve’s saturation knee stole the fit (both preregistered misses, both fixed definitionally on fresh seeds rather than by tuning). The matured detector was then frozen and scored on a held-out labelled benchmark: zero false positives across deceleration, gradual and flat control families at every tested grid density and noise level, onset power 1.00 at the operating resolution of our experiments, and location agreement with a standard change-point method (Methods). Every headline verdict below is additionally stable across 243 adjudication contracts spanning gate, window, truncation, threshold and grid, and—separately, because detector settings and measurement representations are different things—across a preregistered battery of representations of the openness object itself (state coarse-graining, behavioural sampling in place of probabilities, object dualities, reference and truncation changes), whose measured equivalence class, including the two representations that break it and why, is reported in Methods.

With the matured detector, an *onset*-type breakpoint (slow $\rightarrow$ fast) follows the preregistered profile predictions across the tested systems—present where the theory predicts punctuated commitment, absent where it predicts smooth convergence (Fig. 6c). It appears in a committing ant colony, in a learned high-order coordination task (7/8 seeds pooled across the original run and a preregistered five-seed extension), and at the Kuramoto synchronization transition^{24,36}. It is absent—correctly—in an ordinary supervised learner, whose collapse begins at maximum rate and decelerates (a *knee*, not an onset), and it

is unresolvable, and so not claimed, on stored language-model checkpoints whose entropy has already collapsed before the second saved step.

Abruptness scales with system size and criticality

The breakpoint is not a detector artefact but a systematic feature of collective self-amplification. In an ant-commitment model a single chooser *never* shows onset at any gain or grid, while large colonies always do, and the dependence is quantitative (Fig. 6a). Because commitment nucleates from binomial fluctuations of relative size $N^{-1/2}$ that are then amplified at an N -independent rate—a derivation we recorded before running—the commitment time should grow logarithmically while the collapse profile stays fixed. Both predictions hold across $N = 1\text{--}500$: $t_{50} = a + b \ln N$ with $b = 87.6$ (bootstrap 95% CI [41, 124], $R^2 = 0.93$); the closing width is N -invariant (280–290 trips from $N=50$ to 500); and the median collapse curves for $N \geq 50$ fall onto a single universal profile under pure time translation (mean pairwise RMS 0.010 versus 0.266 unaligned)—a sharper statement than equilibrium-style width rescaling⁵. One sub-prediction failed and is reported as such: at the longer preregistered horizon the onset threshold sits between $N = 10$ and $N = 20$, so “onset at every $N \geq 10$ ” was counted a miss rather than the bar being moved. In the Kuramoto system the same instrument tracks the approach to criticality: across coupling $K = 0.9 \rightarrow 2.5$ (10/10 runs onset) the breakpoint time falls monotonically ($6.7 \rightarrow 1.8$, critical slowing down approached from above) and the closing slope rises monotonically ($0.032 \rightarrow 0.199$), while every subcritical run is correctly gated null and the collapse is carried by the pairwise channel in 3/3 seeds (Fig. 6b)³⁷. Abruptness therefore has two independent, manipulable control parameters—system size and feedback strength—both preregistered and both confirmed.

In learned systems, formation and realization dissociate

The most consequential results are in learned agents, because here the possibility space is shaped by optimization rather than fixed physics. We designed a two-phase collective-transport task in which sixteen agents must jointly grip an object before they can push it, structurally delaying the choice of side. Trained with policy gradients³⁸, 10/10 seeds learn the task (success ≥ 0.995) and 10/10 exhibit the predicted signature: the side-openness of the joint policy holds at a plateau for $\sim 17\text{--}19$ steps and then collapses, with an onset breakpoint in every seed ($\Delta\text{BIC } 45.8\text{--}52.7$, t^* concentrated at 16–18; Fig. 2b,d). A preregistered mechanism control—the same task without the gripping phase—shows no such breakpoint (0/5), establishing that it is the structural delay, not learning per se, that produces punctuated collapse (Fig. 2c). The phenomenon reproduces under a different algorithm³⁹ (advantage actor–critic on the byte-identical environment): 5/5 seeds reproduce the plateau-then-collapse shape and the primary hinge ($\Delta\text{BIC } 37.7\text{--}45.5$, $t^* 14\text{--}16$). We report this as a strong partial replication rather than full algorithm independence, because the strict robustness clause (surviving subsample thinning in $\geq 4/5$ seeds) passes in only 3/5—two seeds miss solely on thinned-subsample detector power, and we do not adjust the threshold post hoc.

The same learned system cleanly separates the two senses of emergence that are usually conflated (Fig. 3). Along the *realization* axis—within a single episode—collapse is punctuated (5/5 breakpoints). Along the *formation* axis—across training—the capability appears *smoothly*: outcome-openness expands

($0 \rightarrow \sim 0.65$ within ~ 100 updates, 0/5 breakpoints), and at fine 5-update resolution the success rise is fast but has no structural onset (0/5). Capability formation is smooth and expansive; capability realization is punctuated. This dissociation, observed in the same system with the same instrument, clarifies why debates over whether scaled-model abilities appear “suddenly” have been inconclusive^{11,33}: the capability may form smoothly while each deployment of it commits abruptly.

Punctuated collapse arises without designed barriers

A fair objection is that the grip task’s delayed commitment is imposed by its own state machine, so the flagship functions as a positive control for the instrument rather than a discovery about learning. We therefore preregistered two systems in which *no* environment mechanism delays commitment: every joint regime is available, and all are exactly equivalent, from the first update. In a ten-agent signalling population (five meanings, five symbols, reward only for successful decoding⁴⁰), any of the 120 possible codes could be adopted at any moment; yet convention openness holds at a statistically flat plateau and then collapses with an onset breakpoint in 4/5 seeds (ΔBIC 17.8–30.1), the breakpoint ($t^* = 275\text{--}300$) preceding mutual intelligibility 0.9 (updates 700–1,025) in every onset seed, and the five seeds converge to five *different* codes. In a six-agent division-of-labour task (team reward only when all six interchangeable roles are covered; chance 1.5%), assignment openness shows the same signature in 5/5 seeds and more strongly (ΔBIC 53.6–71.7; closing slope $\sim 30\times$ the plateau slope), again with collapse preceding capability in every seed and five distinct role permutations across five seeds. In both systems the open plateau *is* the joint exploration barrier, and we measure it rather than assert it: mid-plateau, a probe agent unilaterally committing to the *best* available code or role gains essentially nothing (0.011; 0.001), whereas after commitment the cheapest unilateral deviation forfeits the structural maximum of the payoff (0.40 of a 0.40 ceiling; 1.00), and populations from different seeds are mutually incompatible (cross-seed intelligibility 0.14, hybrid-team success 0.05, versus ~ 1.00 within). A code or a role division has value only to the extent that others already share it—and the collapse is nucleated, self-amplifying commitment, the learned analogue of the sharp consensus transitions known from naming-game and convention-formation studies^{41,42}. Neither system depends on its tabular parameterization: replacing the tables with randomly initialized neural policies (per-agent MLPs for the signalling population; one *shared* MLP for all six role agents) reproduces the signature in ten fresh seeds per system—onset in 6/7 learned convention seeds and 10/10 role seeds (ΔBIC up to 162), collapse preceding capability in every one, and every seed converging to a different code or permutation. The neural runs also expose a mechanism the theory predicts: random initialization pre-breaks the symmetry among equivalent regimes, compressing the competition plateau roughly tenfold (commitment within the first 5% of training, which a coarse checkpoint grid provably cannot resolve—the same power floor measured in the detector benchmark), and an initialization-scale sweep moves the commitment time in the predicted direction (monotone in the signalling system; extremes correct but midpoint non-monotone in the role system, reported as a registered miss). This mechanism—seed randomness deciding when and where a run commits—gives a regime-level reading of the finding that emergent language-model capabilities reflect bimodal cross-seed performance distributions rather than deterministic scale thresholds⁴³: each seed crosses the same barrier at a stochastic time, so ensemble averages blur what single-run collapse curves resolve. Punctuated possibility collapse is therefore not an artefact of designed gates or of tabular parameterization: it arose

reproducibly in every tested system where learning must cross an endogenous joint-regime barrier, and the barrier’s width is set by how symmetric the competition starts.

The source typology transfers to learned coordination

Each channel of the ladder is realizable by learning (Fig. 4). A hidden-coordination task with eight individually-observing agents produces a purely *relational* collapse: per-agent marginals stay open (individual entropy ≈ 1.0 bit) while total correlation rises to 6.8/7 bits (Fig. 4a). When the low-order route is blocked with private information, learning builds a genuinely irreducible *higher-order* carrier: the textbook XOR structure emerges with C_{high} 0.94–0.96 bits and pairwise ≈ 0.0004 bits throughout the whole formation history (3/3 seeds; Fig. 4b), and declaring the private cues exogenous re-attributes the identical collapse to C_{env} (Fig. 4c)—the learned version of contract-relativity. Together these give a sharp statement: higher-order structure is not unlearnable but *unfavoured*—gradient learning selects the lowest-order implementation available and builds irreducible structure only when information constraints force it⁴⁴, echoing the difficulty of learning compositional protocols in emergent-communication studies^{45–47} and value-factorization methods in cooperative MARL^{48–50}. Off design, the same ladder correctly reads out Kuramoto oscillators (uncoupled $\rightarrow$ null; common driver $\rightarrow C_{\text{env}}$; single edge $\rightarrow C_{\text{pair}}$), a mechanism vocabulary it never saw. And at matched task performance, the source *profile* separates a learned Overcooked pair⁵¹ from a noise-matched scripted pair through its environment-conditioned component (C_{env} 0.0137 [0.0123, 0.0156] vs 0.0005 [0.0004, 0.0007], non-overlapping; Fig. 4d), even though a single-point coupling measure cannot—the organizational difference is visible in the collapse composition, under the fixed contract, though in no scalar summary.

Remaining openness predicts controllability

If emergence is the closing of a possibility space, then how open that space still is should predict whether the outcome can still be steered. It does (Fig. 5). In the ant system, per-episode intervention flip-rate rises strictly with the episode’s own remaining openness ($0.000 \rightarrow 0.0006 \rightarrow 0.054 \rightarrow 0.205$ across openness bins; Fig. 5d); a closed episode (openness < 0.1) flips 0/2,600 times, and flipped episodes are 0.58 openness units more open than unflipped ones (permutation $p < 10^{-4}$, 8,372 paired counterfactuals). In the learned grip system, a counter-regime impulse switches the outcome with probability 1.0 up to $t = 16$ and only 0.27 by $t = 30$, and side-openness predicts switchability with AUC 0.996 (Fig. 5a,b). This is not a time proxy: holding the intervention time fixed, per-episode openness still separates flippable from locked episodes with AUC 0.974–0.990 at every tested τ (20,480 episodes per cell). Conditioning additionally on the physical order parameter removes the signal—the registered boundary of the claim: in this system openness is an information-sufficient transform of the physical state, so its value is not extra predictive information but that it is computable from the policy alone, without privileged access to the order parameter, and transfers as a decision rule. Openness also works *prospectively*: an intervention rule that simply waits until per-episode openness first drops to a threshold calibrated on the original seeds transfers to five unseen seeds, where it flips the outcome in 99.99% of episodes while acting on average 3.8 steps later than the best fixed schedule (99.6%) and beating random timing by 21 points—remaining openness is a policy-accessible intervention coordinate, not a retrospective correlate, and it acts as late as

each episode allows. A matched-parameter contrast isolates the mechanism: when stances are inertial (a hidden consolidation phase exists), joint openness predicts switchability *better* than the physical order parameter (AUC 0.886 vs 0.849); when the single stickiness parameter is set so no hidden phase exists—all other constants imported unchanged—the advantage reverses (0.811 vs 0.884; Fig. 5c). Openness carries controllability information beyond the order parameter exactly when there is a hidden regime to consolidate.

Where the punctuated form does not appear

The claim is falsifiable and we report its boundary honestly (Fig. 6c). Onset-type collapse is **not** generic to learned optimization. In real deep multi-agent RL (Overcooked, two agents) the learned policies show gradual trajectory-space collapse and increasing role selectivity that track capability but produce no structural onset (0/3 on both a policy-entropy and a trajectory-occupancy object). A registered rescue sequence—smooth population consensus, quorum-threshold populations, and learning-rate scans—also produced no onset (0/20 in the quorum population).

This negative is not left unexplained: the theory names the missing ingredient—cramped-room geometry pins the efficient plan, leaving no equivalent competing joint regimes—and the ingredient can be restored *within the same standard package*. In the official coordination-ring layout, clockwise and counterclockwise circulation are mirror-equivalent joint conventions with value only when shared. With identical PPO mechanics and budget, all eight preregistered self-play seeds end committed to one direction (final direction probability 0.97 or 0.03; six counterclockwise, two clockwise—endogenous symmetry breaking in an unmodified benchmark), while neither cramped-room control ever commits on the same object at the same checkpoint density. The frozen detector certifies a punctuated onset in one seed, with the breakpoint (780k steps) preceding the capability crossing (1M steps), and declines the rest: their convention formation is non-monotone—commitment forms, dissolves and re-forms—which our monotone-collapse instrument was not built to certify. We count the registered onset-rate clause as a miss and report it as an instrument boundary; the regime-level fact that survives every seed is the committed final convention that cramped room never develops. A preregistered within-episode probe locates the commitment level precisely: at mid-training checkpoints (population still split between directions), branched continuation rollouts show that episodes drift with a direction bias but never internally lock (0/5 certified, median within-episode collapse below the 0.1 gate), whereas at the final checkpoints the same probe reads every episode as closed from its first step (initial openness 0.37, one direction, 8/8 seeds) and untrained controls stay open throughout with zero false onsets. The ring convention is therefore a formation-level commitment with no realization stage—the mechanistic complement of the grip system, whose commitment lives entirely within the episode.

A final preregistered experiment tests whether this commitment is causally real rather than a descriptive label. We added unbiased Gaussian noise (two strengths) to stored checkpoints at three training times—very early, at the population’s most open point, and well after commitment—and resumed training for 400k steps with byte-identical mechanics (8 training seeds, 48 continuation runs; the seed is the independent unit and all inference is at seed level). The primary endpoint, declared in the preregistration, is whether the perturbation *disturbs* the commitment (flips the final convention or decommits it): this happens in 7/8 seeds perturbed while open versus 1/8 perturbed after commitment (exact paired sign-flip

test, $p = 0.016$; at run level 8/16 versus 1/16, Fisher $p = 0.008$). Strict direction reversal is a secondary, descriptive endpoint: three open-phase flips and zero late flips at run level, a contrast in the predicted direction that does not reach significance across eight seeds ($p = 0.13$) and which we do not lean on. The stored openness at the perturbed checkpoint predicts strict flips across the pooled 48 runs (AUC 0.85, preregistered threshold 0.70; descriptive, as runs cluster within seeds; four of the five flips occurred at openness ≥ 0.90), and capability recovers to 0.92 of the unperturbed rate after late perturbation—the same perturbation-plus-retraining budget that can no longer move the convention is amply sufficient to relearn the task. What locks at commitment is the institution, not the learning. Two boundaries of this claim are explicit. The experiment validates the functional distinction between *open* and *committed* policy states in an unmodified community benchmark; it does not certify a breakpoint time t^* as a causal transition point, because formation here is non-monotone and only one seed carries a certified onset. And because open and late checkpoints differ in training time as well as openness, the timing contrast alone cannot exclude generic training maturity; the fixed-time analysis that rules out the time confound exists in the grip system, not yet here. Among classic cases the picture is deliberately mixed: Vicsek flocking⁵², Swift–Hohenberg pattern selection⁵³ and Schelling segregation⁵⁴ organize strongly but collapse *gradually* under our object, while the Potts model⁵⁵ reproduces the known continuous-versus-first-order distinction through hinge strength and hysteresis. The honest conclusion is not “everything is punctuated collapse” but a classification: possibility-collapse profiles sort systems into punctuated, gradual, parameter-axis and early-warning forms, and onset appears specifically when the system must cross a joint-regime barrier—whether that barrier is physical (the grip gate), informational (the blocked low-order channel) or endogenous to learning itself (a convention or role structure that has value only once shared). Dense-reward optimization with no such barrier collapses gradually, and our instrument says so.

Discussion

We have argued, and measured, that emergence can be defined on the object that actually reorganizes—the effective joint possibility space—rather than on the observable of convenience, and that this choice avoids the specific failure mode—discontinuous scoring of continuous competence—behind the metric-artefact problem that has dogged emergent-ability claims³³: amplitude and abruptness are provably separate axes, and a discontinuous scoring function produces zero collapse under our instrument. The definition is generative rather than nominal. It supplies a source typology that transfers from analytic ground truth to physics to learned agents; a breakpoint whose presence and absence track the theory’s own predictions across ant colonies, a synchronization transition and learned high-order coordination; two systematic and manipulable dependences of abruptness on system size and coupling; and a control-theoretic payoff, since remaining openness predicts whether an intervention can still flip the outcome.

For machine intelligence specifically, the formation/realization dissociation clarifies a persistent confusion. “Emergent abilities” of scaled models¹¹ and the within-run appearance of a coordinated strategy in self-play agents^{13,56,57} are different events on different timescales: our learned flagship shows the first as smooth and expansive and the second as punctuated, in the *same* system. This reframes the debate about whether capabilities “suddenly” appear: the capability may form smoothly while

each deployment of it commits abruptly. Concurrent work detects higher-order structure in multi-agent language-model groups through partial information decomposition of time-delayed mutual information⁵⁸; that programme asks whether synergy is *present*, whereas ours asks when the joint possibility space *closes*, how abruptly, through which channel, and whether the closure can still be steered—the two are complementary, and the collapse object adds the timing and control dimensions that presence tests do not address.

The limitations are real and stated as scope, not hidden. Onset-type collapse is not generic to deep MARL at current scale: dense-reward Overcooked optimization collapses gradually. Our designed systems (the grip gate; the blocked-channel XOR task) are mechanism-isolating controls, and we now know they are not load-bearing: convention formation and role lock-in reproduce the punctuated signature with no designed barrier at all. What separates the two classes, on present evidence, is the presence of a joint exploration barrier, and the coordination-ring experiment tests exactly this boundary inside the standard benchmark that produced the negative: restoring equivalent competing regimes restores committed conventions in every seed. Certified onsets remain rare there because that system’s commitment is non-monotone—an instrument boundary we state rather than engineer away (a preregistered settled-envelope extension failed its own synthetic validation and was retired unused; Methods). Certifying non-monotone commitment, and extending to large-scale self-play and open-ended environments, are the clear next steps. We do not claim a causally verified commitment window—a one-pulse-destroys-the-regime hypothesis failed its own falsification clause, and formation proved re-entrant, which we report as a persistence property. We make no breakpoint claim for stored language-model checkpoints, whose grids are too sparse to resolve onset. We do not claim the openness object is representation-independent in the absolute: the audit delivers a measured equivalence class—verdict and location survive object dualities, behavioural sampling and state coarse-graining down to the detector’s resolution floor, and break when a representation erases the latent channel—not blanket invariance. And the universal equivalence “emergence = onset breakpoint” remains unproven: several canonical cases collapse gradually. What we offer is narrower and, we believe, more useful—a calibrated, preregistered, falsifiable instrument that measures *how*, *when* and *through which channel* a possibility space closes, and that earns its claims by reporting, in the open, every place it says no. A natural next step is to connect the collapse object to internal-representation geometry in trained networks⁵⁹, extending the instrument from behaviour to mechanism.

Methods

Possibility space and openness. For a system of n parts we define the effective joint possibility distribution P_t over the relevant state–action–trajectory support at analysis time t , estimated from rollouts or exact enumeration where tractable. Openness is $O_t = H(P_t)/H(P_{\text{ref}})$ with H the Shannon entropy in bits and P_{ref} the maximum-entropy reference on the same support; collapse is $C_t = 1 - O_t$ and effective possibility count is $N_{\text{eff}} = 2^{H(P_t)}$. Entropies are estimated with a Miller–Madow-style bias correction and, for the joint objects, with the nested ladder below rather than by direct high-dimensional estimation.

Source decomposition. The nested maximum-entropy ladder constrains, in order: (i) the statistics of the declared exogenous variables and conditional independence given them (environment-conditioned); (ii) additionally

each part’s marginal (individual); (iii) additionally all pairwise marginals, following the connected-correlation construction of Schneidman et al.³⁵ (pairwise); and (iv) the full joint. C_{env} , $C_{\text{individual}}$, C_{pair} and C_{high} are the successive entropy reductions. Three properties matter for interpretation. First, each stage only adds constraints, so entropy is non-increasing along the ladder and every component is non-negative by construction. Second, the ladder is a sequence of information projections defined for *every* joint distribution—it is not a parametric model of the data—so “model misspecification” does not arise; a generator that mixes sources simply deposits entropy reduction at several levels at once, which is the intended reading. Third, the stage order is not a free knob: given a declared exogeneity boundary, the individual-to-pairwise-to-higher progression is the canonical hierarchy of marginal constraints, and the only genuinely movable choice—the boundary itself—is exactly the declared contract-relativity demonstrated in Fig. 4c. The provenance boundary is fixed before analysis in every experiment. A preregistered stress audit quantifies the remaining freedoms. Off-family generators the ladder was never built from are attributed correctly (a modular-sum triple with exactly independent pairwise marginals loads 3.32 bits on C_{high} with $C_{\text{pair}} = 0$; a Markov copy chain loads purely on C_{pair} with $C_{\text{high}} = 0$; fifty random Dirichlet joints give zero negative components and an exact telescoping identity). The constraint sets form a filtration, so no individual/environment reordering exists; the one genuine order freedom—declaring the environment before or after the pairwise stage—leaves the individual and higher-order components exactly invariant and moves the environment/pairwise split by at most 0.26 bits across an 81-cell grid. Finite-sample behaviour is measured, not assumed: median absolute component error is ≤ 0.018 bits at 3×10^4 samples for the three-agent test system, with no negative estimates in 120 runs, and the small-sample bias concentrates in C_{high} (0.31 bits at $n = 300$), which sets an explicit sample floor for higher-order claims. One registered audit prediction missed: when generator knobs interact strongly (pairwise copying under a parity mechanism), realized structure genuinely relocates across orders—copying makes the parity constraint first-order—so the ladder attributes it to the individual level. We report this as a property, not a defect: the ladder measures realized distributional structure, not generator labels.

Breakpoint detector. Collapse curves are fit with one- and two-regime piecewise-linear models; the breakpoint is accepted when the two-regime ΔBIC exceeds 10 and survives parity-thinning of the checkpoint grid, the curve passes an effect-size gate (max drop ≥ 0.1), and the analysis window is truncated at saturation (first point within 5% of the final value). An *onset* type requires $|\text{slope}_{\text{after}}| > |\text{slope}_{\text{before}}|$; the reverse ordering is a *deceleration* knee. The gate and saturation-truncation were added in response to two preregistered control failures (a flat subcritical false positive; a saturation-knee capture) and re-tested on fresh seeds.

Detector validation. Because the detector was matured through its own registered failures, it was frozen and then scored on an independent, preregistered synthetic benchmark of 200 curves per family (onset, deceleration knee, linear gradual, sigmoid, flat) per condition, with labels fixed before running. At the reference operating point (80 grid points, noise $\sigma = 0.02$) onset power is 1.00 and the false-positive rate across all control families is 0.000; power rises monotonically with grid density (0.00/0.62/0.995/1.00 at 12/20/40/80 points) while the false-positive rate remains 0.000 at every density, and degrades gracefully with noise (0.93 at $\sigma = 0.08$, false positives 0.000 throughout). Zero observed false positives among 200 curves per control family bounds the true rate at $\leq 0.5\%$ per condition (95% rule-of-three over the pooled 600 controls); we report the bound rather than claiming the rate is exactly zero. The detector adjudicates monotone collapse only: a preregistered extension to non-monotone commitment (a settled-openness envelope recording irrevocable closure) kept false positives below 0.05 in every validation round but never exceeded 0.71 power against its required 0.90 on synthetic non-monotone positives, and under its own preregistered stopping rule was closed without ever being applied to system data; certifying non-monotone regime formation therefore remains an open instrument problem, which is why the coordination-ring onset count is reported at 1/8. A semi-synthetic injection audit quantifies the matching noise floor: commitments injected at known times into the ring pipeline’s own estimator (real grids, real committed-episode counts, binomial direction noise) are recovered with false-positive rate 0.01 and median t^* error 1% of span, but detection power

saturates at 0.88 (registered threshold 0.90, recorded as a near-miss)—the 30-episode evaluation budget itself caps single-curve power below the synthetic-library ceiling. Breakpoint locations agree with an independent standard change-point method (binary segmentation, RBF cost) to within 10% of span on onset curves; unlike generic change-point detection, the instrument correctly declines to call onset on deceleration and gradual curves. The zero power at 12 grid points is retained as an honest floor: it is why no claim is made on sparse language-model checkpoint grids.

Adjudication-contract robustness. All headline breakpoint verdicts were re-adjudicated across 243 analysis contracts per system, crossing saturation fraction $\{0.02, 0.05, 0.1\}$, analysis window $\{0.75, 0.875, 1.0\}$ of the curve, effect-size gate $\{0.05, 0.1, 0.15\}$, ΔBIC threshold $\{8, 10, 12\}$ and checkpoint-grid stride $\{1, 2, 3\}$. For the grip flagship, the committing ant colony and the learned high-order task the onset verdict is unchanged in 100% of adequate-resolution cells and the breakpoint location varies by 1%, 8% and 11% of curve span respectively; aggressive grid coarsening reduces detectability (a power effect quantified by the validation benchmark above), never the location. The collapse object itself is fixed by the theory (the joint commitment space), not chosen by fit: the raw action-entropy of the grip system *rises* during the same window and is reported as such.

Observation contract. Every entropy is computed relative to a declared contract with six components: the variables admitted to the joint space, the trajectory horizon, the discretization, the reference distribution (maximum entropy on the same support), the probability-truncation rule, and the provenance boundary separating environment from system. Two properties anchor the contract’s objectivity. First, an invariance that holds exactly: O_t is unchanged under any bijective recoding of states or actions, because Shannon entropy is permutation-invariant and the reference is defined on the same support—openness cannot be manufactured or destroyed by renaming. Second, an admissible class that is measured rather than assumed: representations remain admissible while they separate the competing regimes at or above the detector’s resolution floor, and the equivalence battery below maps where that class ends (non-injective coarse-graining that merges regime-distinguishing states, or truncation that deletes the latent channel, breaks the verdict—mechanistically, not capriciously). The contract is frozen before adjudication and the instrument refuses outside it. One component deserves emphasis: the framework does not discover the regime variable from raw trajectories; it certifies a *declared* regime object. In every system here that object was named before analysis with a domain justification—side choice (bridge), grip configuration and push side (transport), code mapping (signalling), role permutation (division of labour), XOR carrier (blocked channel), circulation direction (Overcooked ring), phase locking (Kuramoto), magnetization sector (Potts)—and where no credible regime object exists the correct output is refusal, not a verdict. The instrument is thus a general measurement contract for declared collective regime objects, not an automatic emergence detector for arbitrary raw systems. The analyst’s freedom is thereby the same freedom every measurement has—the choice of ensemble in thermodynamics, of coordinates in mechanics—made explicit, fixed in advance, and accompanied by its measured range of validity.

Representation equivalence. Detector settings and measurement representations are different freedoms, so they were audited separately. On byte-identical retrained seeds we recomputed the openness object under twelve preregistered representations—for the convention system, the population-mean mapping, per-agent mean entropy, the listener-side dual, a 2,048-sample behavioural estimate, a checkpoint-zero empirical reference, $\varepsilon = 0.01$ truncation and $5 \rightarrow 3$ symbol coarse-graining; for the grip system, policy probabilities, state quantization at 0.1 and at 0.5, realized 16-agent action counts, and $\varepsilon = 0.01$ truncation—and adjudicated each with the frozen detector. The onset verdict survives in 8/12 cells and the breakpoint location is essentially representation-invariant wherever a verdict exists: $t^* = 250\text{--}300$ of a 4,000-update span (1.25%) in the convention system, and identical ($t^* = 17$) in every preserving grip representation. The registered $\geq 90\%$ preservation bar was missed, so per the registered clause we report the measured equivalence class instead of claiming blanket invariance. All four breaking cells

fail for identifiable reasons and still locate the commitment at $t \approx 19\text{--}22$: two fall just below the conservative $\Delta\text{BIC} = 10$ bar while agreeing on location (per-agent entropy, which is an individual-level object rather than the joint convention; $5 \rightarrow 3$ binning); 0.5-level state quantization compresses the transition into a one-to-two-checkpoint cliff that fails the frozen thinning clause (a resolution limit, not a disappearance: the unthinned fit gives $\Delta\text{BIC} 111$, onset-type); and ε -truncation zeroes the side channel while it is latent ($<1\%$ of action mass during the grip phase), making the object non-monotone—truncating small probabilities destroys precisely the still-open possibilities the object exists to measure. The equivalence class is therefore: verdict and location survive object dualities, behavioural sampling, reference changes and state coarse-graining down to the detector’s resolution floor; they break only when the representation erases the latent channel or compresses the transition below that floor.

Emergence certificate. The frozen instruments are packaged into a single standardized adjudication (`emergence_certificate`) so that a new system receives a reproducible verdict: a three-condition qualification (regime-level: the collapse gate *and*, where the source analysis is available, joint-beyond-marginal reorganization; endogenous: the declared provenance boundary; persistent: no re-opening beyond 0.1 of reference within the observed window), the intensity profile as a vector (amplitude, abruptness class, sharpness, t^* , source ranking), and a categorical verdict with an explicit *unresolvable* class below the measured power floor and a low-power flag between 12 and 40 grid points. We deliberately do not output a scalar “emergence score”: amplitude and abruptness are demonstrated independent axes, and any fixed weighting of them would reintroduce the observable-of-convenience failure this framework exists to remove. Applied to the paper’s systems, the certificate issues “emergent: punctuated” for the grip, convention, role and $N=100$ ant systems; the single-ant control—with nearly the same amplitude (0.92) as the colony—correctly *fails* qualification, because its collapse is purely individual-source rather than a joint regime, and the Overcooked occupancy object falls below the collapse gate with a low-power flag. Same entropy drop, different verdicts: the discrimination is exactly what a bare entropy measure cannot provide.

Learned systems. The grip-transport flagship uses 16 agents, grip threshold 6, a two-phase grip-then-push dynamics (grip gain 0.06, decay 0.01), trained by REINFORCE³⁸ (1,200 updates, batch 512, learning rate 2×10^{-3}) and, for the robustness check, by advantage actor–critic³⁹ on identical environment parameters. The hidden-coordination flagship uses 8 individually-observing agents with per-agent stances; the sticky variant sets stance-change probability 0.25 and the matched control sets it to 1.0 with all other constants imported unchanged. Overcooked runs use the standard cramped-room layout⁵¹. The non-constructed convention system is a ten-agent Lewis signalling population⁴⁰ (five meanings, five symbols, tabular policies, REINFORCE, 512 random speaker–listener pairs per update, 4,000 updates); the role system is six agents choosing among six roles with team reward only for full coverage (6,000 updates, otherwise identical training); in both, openness is computed from policy probabilities at 25-update checkpoints and adjudicated by the frozen detector. The neural replications keep both environments and recipes fixed and replace the tables with one-hidden-layer networks (32 tanh units, Adam 3×10^{-3} , ten fresh seeds per system): per-agent speaker and listener MLPs for the signalling population, and a single MLP shared by all six role agents, so that role differentiation must emerge inside one network. Because random initialization compresses commitment into the first few hundred updates, the neural runs are evaluated at 5-update resolution—a preregistered measurement-density amendment justified by the detector’s validated grid-power curve; at the original 25-update grid the onsets are under-resolved (1/7 and 1/10), which is recorded alongside the fine-grid result. The initialization sweep varies the input-weight scale $\sigma \in \{0.02, 0.1, 0.45\}$ at five seeds per cell. The coordination-ring study trains standard `overcooked_ai` self-play PPO (mechanics and budget identical to the cramped-room runs, 2M steps) on the official `coordination_ring` layout, eight seeds, checkpoints every 20k steps; the regime object is the entropy of the pair’s circulation direction (net winding around the central counter, Laplace-smoothed over 30 evaluation episodes per checkpoint), declared before running. A parallel preregistered attempt on the unmodified MPE `simple_spread` benchmark did not reach its frozen competence precondition (conflict-episode coverage 3% after 3,000 PPO updates) and is recorded as a training

failure with no theory claim made either way. The causal commitment test perturbs stored ring checkpoints at 100k steps, at the seed’s most-open checkpoint (last with direction probability in $[0.3, 0.7]$ and ≥ 20 committed evaluation episodes), and at 1.6M steps, adding per-tensor Gaussian noise of 0.25 or 0.5 relative standard deviations, resuming training for exactly 400k steps with unchanged mechanics and schedule position, and scoring the final convention against the seed’s unperturbed direction; all times, scales, margins and clauses were frozen in the protocol file before any run. Because the two noise scales share a training seed, continuation runs cluster within seeds; the run-level Fisher test was the frozen clause, and a stricter seed-level paired analysis (exact sign-flip permutation over the eight seeds, labeled post-hoc in the protocol file) confirms the primary endpoint and demotes the strict direction-reversal contrast to descriptive. The finite-size study uses sizes $N \in \{1, 2, 5, 10, 20, 50, 100, 200, 500\}$, 30 episodes per size and a 1,500-trip horizon, with t_{50} and width read from median openness crossings and the log-law slope bootstrapped over episodes. The ant, Kuramoto, Potts, Vicsek, Swift–Hohenberg and Schelling models follow their standard definitions (Supplementary Methods).

Barrier measurement. The joint-exploration-barrier claim is tested by deterministic replay of both learned positive systems with policy snapshots at every checkpoint (replay validity: all ten re-run seeds reproduce the stored final codes and role assignments exactly). At the checkpoint nearest $t^*/2$ (onset seeds only), the unilateral adoption gain of agent i is the payoff of a probe hard-committed to the best of all $K!$ codes (respectively all R role locks) against the unmodified population, minus the payoff of i ’s own current policy, averaged over agents; payoff is symmetric expected intelligibility with a random partner (convention) or the permanent of the row-stochastic role matrix (roles). At the final checkpoint the unilateral deviation cost is the payoff on the population’s majority regime minus the best payoff over all alternative committed regimes. Cross-play pairs speaker populations with listener populations from different seeds (20 ordered pairs) and assembles hybrid role teams from every proper subset split of every distinct-assignment seed pair (620 teams). All four outcome clauses were frozen before the run; one threshold-calibration miss (the convention deviation-cost clause set 0.50 where the payoff structure caps the cost at $0.4 \times$ within-population payoff for $K=5$ permutation codes, the measured value 0.3996 sitting at that ceiling) is reported verbatim in the protocol file. What these quantities measure is a *unilateral coordination barrier*—the absence of single-agent payoff gradient toward any regime, and the single-agent cost of leaving one—not the full performance landscape over joint policy paths; multi-agent coordinated escape routes are not enumerated.

Scalability. The pipeline’s cost is dominated by openness estimation, not by the detector (which fits two linear models per gridpoint and is linear in grid length). Openness requires the entropy of the declared possibility object, estimated either exactly from policy probabilities (learned tabular and neural systems), or from empirical distributions over the declared coarse-graining (physical systems), whose sample complexity we measured directly: in the finite-sample audit the median absolute error of every ladder component is ≤ 0.02 bits at 3×10^4 samples, rising at 300 samples to 0.09 bits (individual), 0.31 (higher-order) and 0.81 (pairwise), so the pairwise term is the first casualty of scarce data. Total openness itself is linear in population size times horizon, and the breakpoint stage ran unchanged at population sizes up to $N=500$ in the finite-size study; the source decomposition is the only superlinear stage (pairwise marginals cost $O(N^2)$ and the higher-order maximum-entropy fit is the practical ceiling), so qualification and profile can always be computed at scales where the full typology cannot. All experiments in this paper run on CPU except the Overcooked PPO training; no single run exceeds a day of wall-clock time.

Preregistration and integrity. Every predicted outcome was frozen, with an explicit falsification clause, in an internal protocol file before the corresponding run; passes and misses are reported identically, and each amendment is recorded in place with its trigger. We characterize these as internally time-stamped frozen protocols rather than third-party registrations: they were not lodged with an external registry, and we release them verbatim—including every miss, every failed instrument extension and every post-hoc analysis labeled as such—together with the code, raw outputs and an output manifest, so that the ordering of protocol and result can be audited from the released

record. All numbers in the main text and figures are produced by a single extraction script run directly against the raw output files; no value is transcribed by hand. One float32 numerical defect (an entropy clamp that rounded to a degenerate value) was found, fixed in code, and both affected experiments were rerun with identical seeds, with the corrupted outputs archived. No thresholds were changed after seeing results.

Reporting summary. Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article. The use of large-language-model agents is disclosed in the dedicated section in the back matter.

Data availability

The raw output of every experiment (one JSON file per registered run), together with the frozen protocol files that fix each prediction and falsification clause before running, is provided in the accompanying repository and will be archived with a permanent DOI on Zenodo upon acceptance. Source data for each main-text figure are generated directly from these files by the plotting script; no intermediate hand-edited data are used.

Code availability

All simulation, training, analysis and figure-generation code is available in the accompanying repository and will be archived on Zenodo upon acceptance. The manuscript's quantitative claims are reproduced by a single extraction script (`manuscript_numbers.py`) that reads the raw outputs and prints every number cited in the text and figures, and by a figure-generation script (`make_figures.py`) that draws all panels from the same files.

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Author contributions

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Competing interests

The authors declare no competing interests.

Use of large language models

Large-language-model agents were used throughout this work, under continuous human direction and review, to design and implement experiments, refine analysis protocols, write and debug code, and draft and edit text. All predicted outcomes, falsification clauses and adjudication thresholds were fixed in the protocol files before the runs they govern; every number reported in the text and figures is generated by the released code from raw experiment

outputs and cross-checked by an automated extraction script. The authors independently reviewed and validated the code, protocols and analyses, and take full responsibility for all claims.

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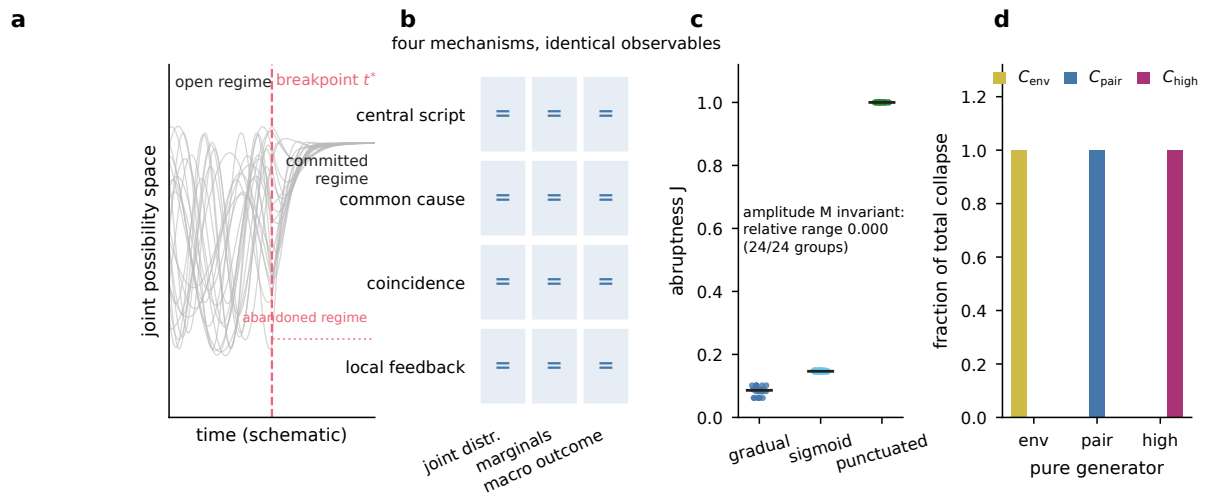


Figure 1. The problem and the instrument. **a**, The intuition: many parts wander in a wide joint possibility space (grey) until a breakpoint t^* after which trajectories commit to one regime and abandon the alternatives (schematic). **b**, Four generators—central script, common cause, coincidence and local feedback—constructed so that no statistic of the shared joint-action distribution separates them (identical joints, marginals and macroscopic success). What separates them is the declared contract: conditioning on declared exogenous variables, interaction-severing counterfactuals and persistence checks assign each to its predicted certificate component—attribution is contract-relative, not mechanism discovery from outcome statistics. **c**, Full-factorial calibration (72 cells): amplitude M is invariant across temporal shape while abruptness J strictly orders punctuated > sigmoid > gradual. **d**, Source-decomposition ladder on analytic ground truth: each pure generator loads almost entirely onto its own component.

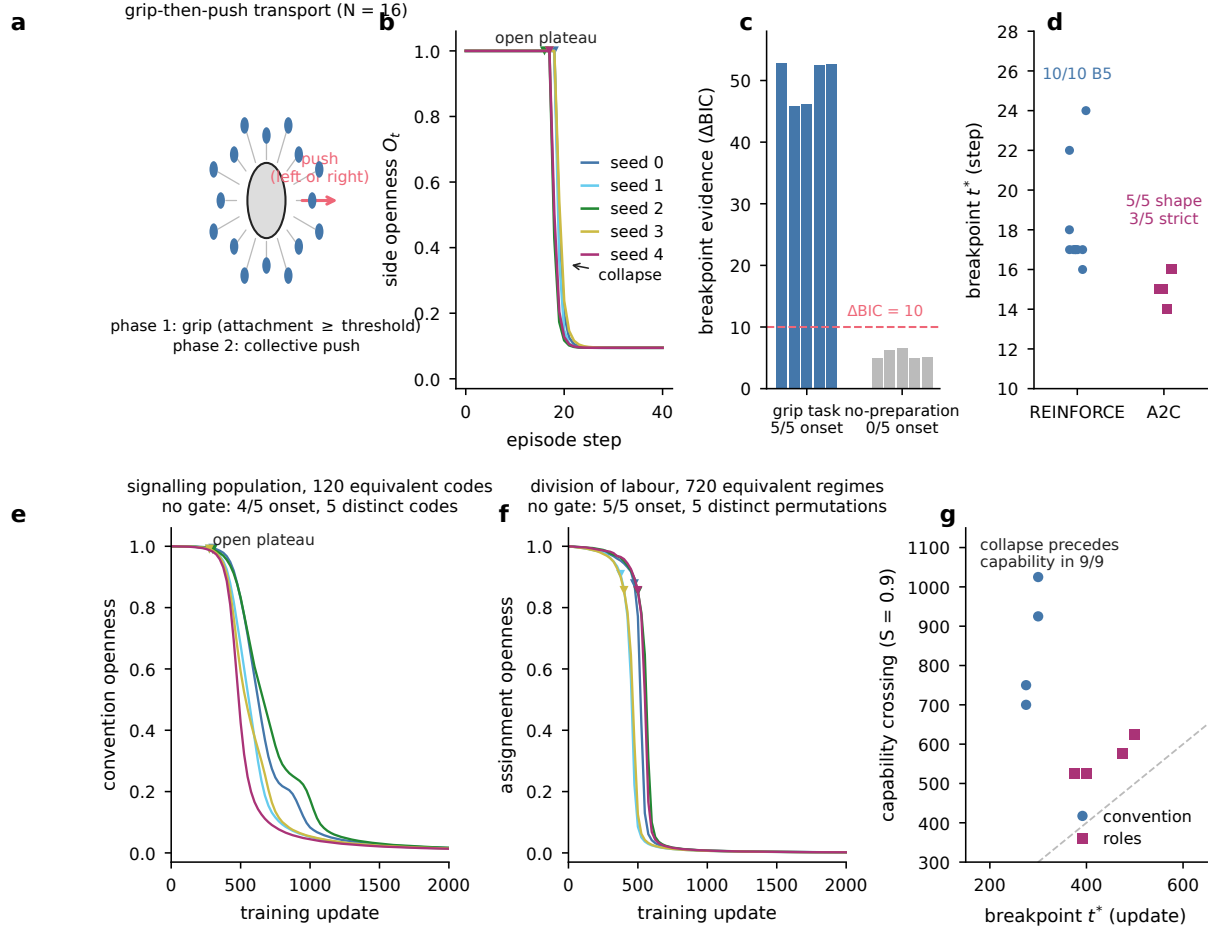


Figure 2. Punctuated realization in a learned collective. **a**, The grip-then-push transport task ($N = 16$): agents must jointly grip before the object can be pushed, structurally delaying the choice of side. **b**, Side-openness of the joint policy holds on a plateau and then collapses; triangles mark the fitted breakpoint t^* (5 REINFORCE seeds). **c**, Mechanism contrast: the grip task shows a breakpoint in 5/5 seeds (ΔBIC well above 10) whereas a matched no-preparation task shows none (0/5). **d**, Breakpoint location across seeds and algorithms: 10/10 for REINFORCE, and a strong partial replication under A2C (5/5 shape and primary hinge, 3/5 strict). **e**, Non-constructed positive I: a ten-agent signalling population with 120 exactly equivalent codes and no gate holds a flat open plateau, then commits with an onset breakpoint (4/5 seeds; triangles), each seed to a different code. **f**, Non-constructed positive II: six agents locking into a division of labour among 720 equivalent role permutations (5/5 onset, five distinct permutations). **g**, In all nine onset seeds the possibility-space breakpoint precedes the capability crossing ($S = 0.9$): commitment is measurable before the ability is expressed.

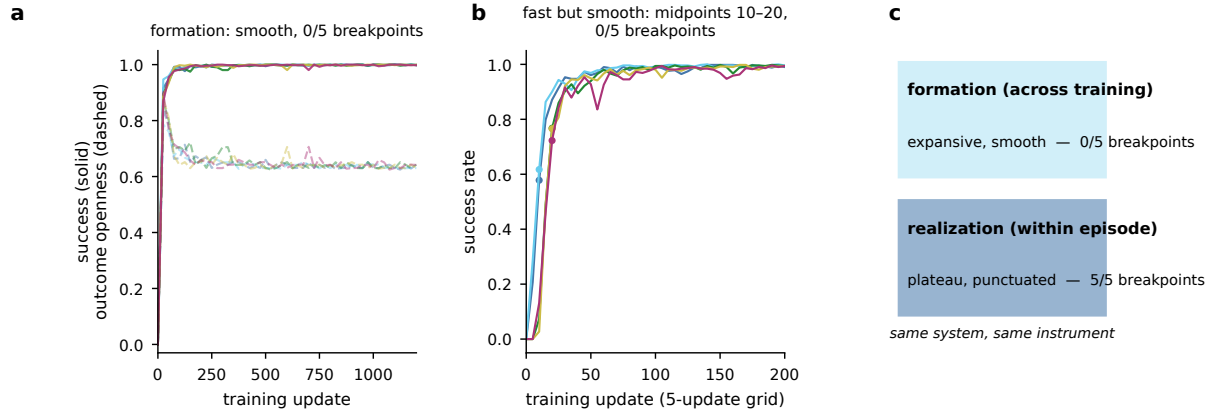


Figure 3. Two timescales dissociate in one learned system. **a**, Formation axis (across training): success (solid) and outcome openness (dashed) rise smoothly, no breakpoint (0/5). **b**, At fine 5-update resolution the success rise is fast but still has no structural onset (0/5); dots mark per-seed midpoints. **c**, Summary: capability formation is smooth and expansive; capability realization is punctuated—same system, same instrument.

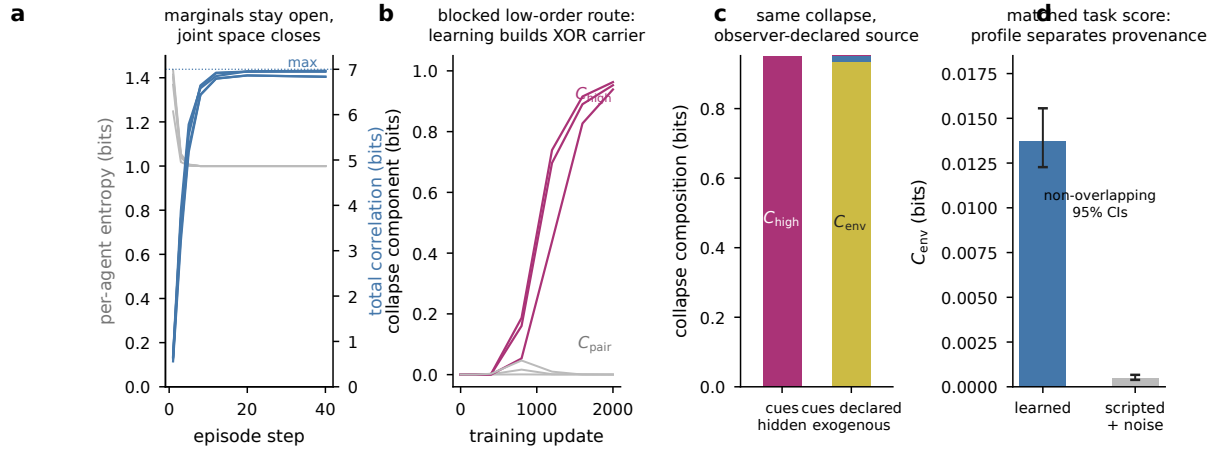


Figure 4. The source typology transfers to learning. **a**, Hidden-coordination task: per-agent entropy stays open while total correlation rises to near the 7-bit maximum—a purely relational collapse. **b**, With the low-order route blocked, learning builds an irreducible higher-order (XOR) carrier: C_{high} grows while C_{pair} stays near zero. **c**, Contract-relativity on the learned system: declaring the private cues exogenous re-attributes the identical collapse from C_{high} to C_{env} . **d**, At matched task score, the environment-conditioned component separates a learned pair from a noise-matched scripted pair (non-overlapping 95% confidence intervals).

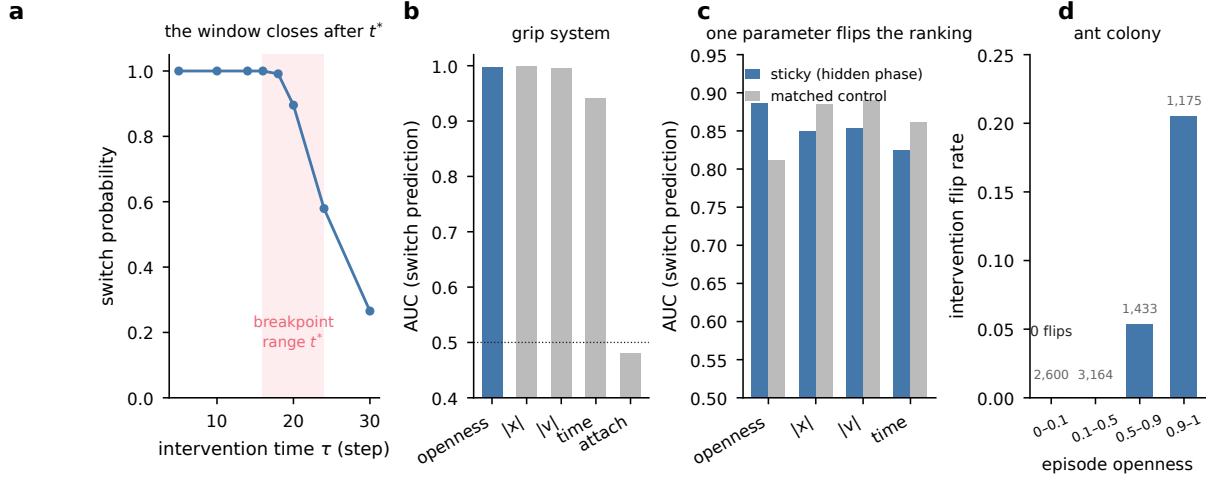


Figure 5. Remaining openness predicts controllability. **a**, Grip system: probability that a counter-regime impulse switches the outcome, as a function of intervention time; the window closes after the breakpoint range (shaded). **b**, Openness predicts switchability as well as or better than physical baselines (AUC). **c**, Matched-parameter causal contrast: openness beats the order parameter only when a hidden consolidation phase exists (sticky), and the ranking reverses in the control. **d**, Ant colony, paired counterfactuals: intervention flip-rate rises with the episode’s own remaining openness; closed episodes never flip.

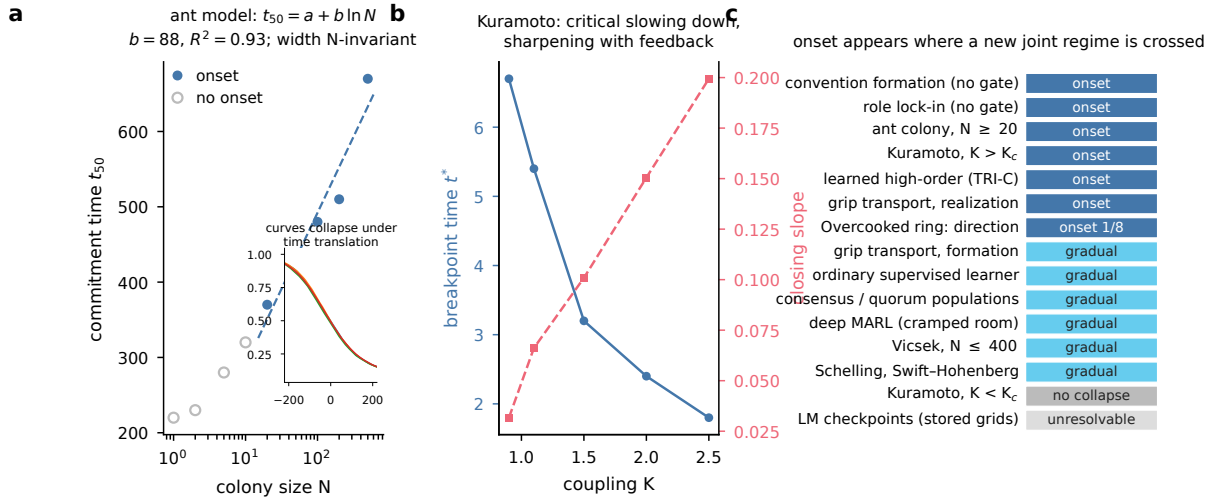


Figure 6. Scaling and scope. **a**, Finite-size scaling in the ant model: commitment time is consistent with $t_{50} = a + b \ln N$ ($b = 88$, 95% CI [41, 124], $R^2 = 0.93$; dashed fit), as predicted by nucleation from $N^{-1/2}$ fluctuations; open circles, sizes below the onset threshold. Inset: median collapse curves for $N = 50$ –500 superpose under pure time translation (aligned RMS 0.010 vs 0.266 unaligned)—the profile is universal, only the timing shifts. **b**, Kuramoto criticality: breakpoint time falls and closing slope rises as coupling increases; subcritical runs are gated null. **c**, Classification: onset appears where a joint-regime barrier is crossed—including without any designed gate (convention formation, role lock-in) and in the standard Overcooked coordination-ring layout, where all 8 seeds commit to a circulation direction and 1/8 certifies onset (Results)—and is absent for gradual convergence (formation, ordinary learners, consensus/quorum populations, cramped-room deep MARL, several classic cases).